Name ____________________ Section ________

MTH:2310 DISCRETE MATH
QUIZ 5
DUE: THURSDAY OCT. 16

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: *(4 points)*
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. *(12 points) Sequences and Summations*
Find the first five terms of the following sequences. Recall $\lfloor x \rfloor$ is the floor function.

(a) *(4 points)* $\left\{ 2^n + \left\lfloor \frac{n+1}{2} \right\rfloor \right\}_{n=0}^{\infty}$

(b) *(4 points)* $\left\{ \left(\frac{-1}{n+1} \right)^{n-2} \right\}_{n=1}^{\infty}$

(c) *(4 points)* $f_1 = -1$,$f_2 = 2$, $f_3 = 4$ and $f_{n+1} = f_n + f_{n-1} + f_{n-2}$ for $n \geq 3$

Problem 2. *(8 points) Sequences and Summations*
Use an expression for the n^{th} term to show that each of the following sequences are solutions of the following recurrence relation $a_n = 3a_{n-2} + 2a_{n-3}$.

(a) *(4 points)* $\left\{3n(-1)^{n-1}\right\}_{n=1}^{\infty}$

(b) *(4 points)* $\{-5(2)^n\}_{n=2}^{\infty}$

Name ______________________________ Section ________

Problem 3. *(6 points) Recursive Definitions*

Define a recurrence relation which produces the following sequence.

$$\{a_n\}_{n=0}^{\infty} = \{-4, -3, -1, 5, 29, 149, ...\}$$

Problem 4. *(6 points) Recursive Definitions*

Let $f : \mathbb{N} \to \mathbb{Q}$ be a function where $f(0) = 1$ and $f(n+1) = \left(\dfrac{f(n)}{f(n)+2}\right)^n$ for all $n \geq 0$. Find $f(1)$, $f(2)$, and $f(3)$.

Problem 5. *(8 points) Recursive Definitions*
Let S be a such such that $-4 \in S$ and for any $x, y \in S$, $x - y \in S$.

(a) *(4 points)* Find at least 7 distinct members of the set S.

(b) *(4 points)* Explicitly define S using set-builder notation.